

Performance & Best Practices

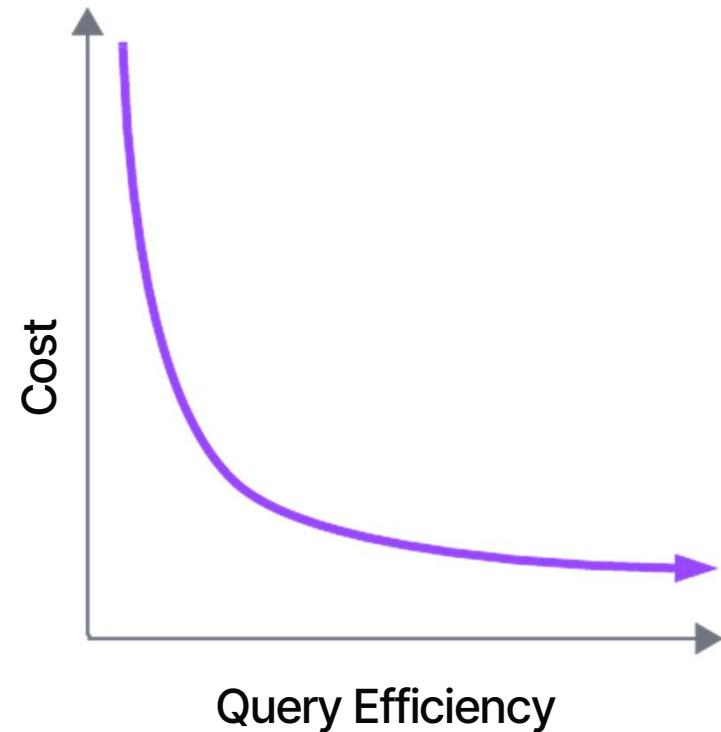
Lesson objectives

Performance & Best Practices

1. Describe at a high level how a query is executed on the cluster.
2. Explain the function of a query plan.
3. Outline tips for efficient querying.
4. List common pitfalls to avoid.

Why Performance Matters

- Analysts spend time waiting on slow queries.
- Inefficient queries = higher costs.
- Optimized queries = faster insights + lower compute.



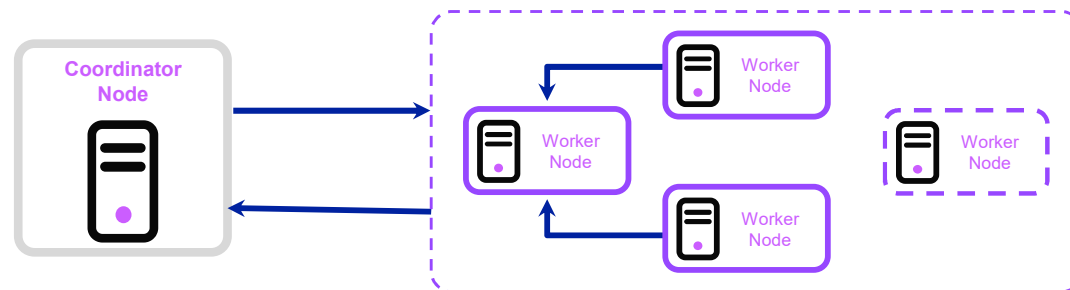
Server stereotypes

Coordinator node

Server that is responsible for parsing statements, planning queries, and managing Trino worker nodes.

Worker nodes

Server that is responsible for executing tasks and processing data. Worker nodes fetch data from connectors and exchange intermediate data with each other.

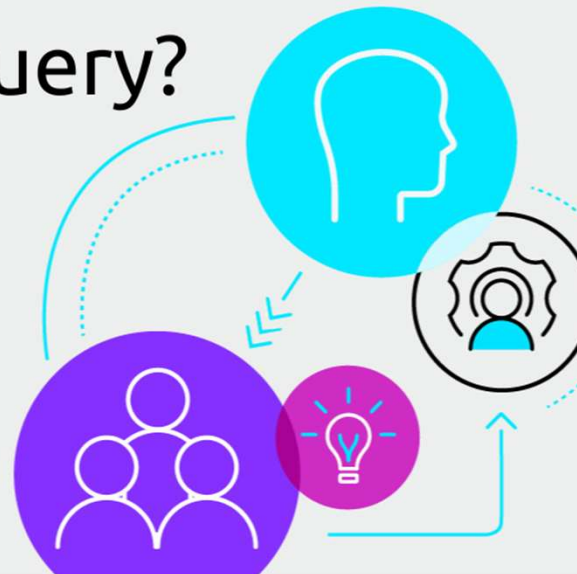


How a Query is Executed

Watch Together – Interactive Discussion / Q&A

How does Trino process a query?

 Starburst Academy



Understanding Query Plans

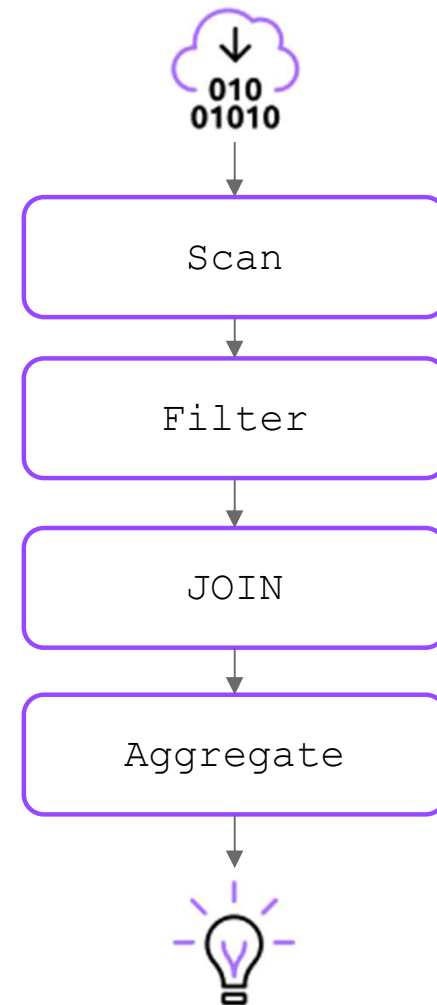
At a High Level

Starburst translates SQL into a query plan.

A plan shows:

- How data is read (scans).
- How it's joined or aggregated.
- Where bottlenecks may appear.

Key takeaway: You don't need to be an engineer — just know that query design impacts the plan.



Tip #1: Filter Early

- Only select columns you need.
- Always use `WHERE` clauses to narrow data.
- Push filters down to the source when possible.

Example:

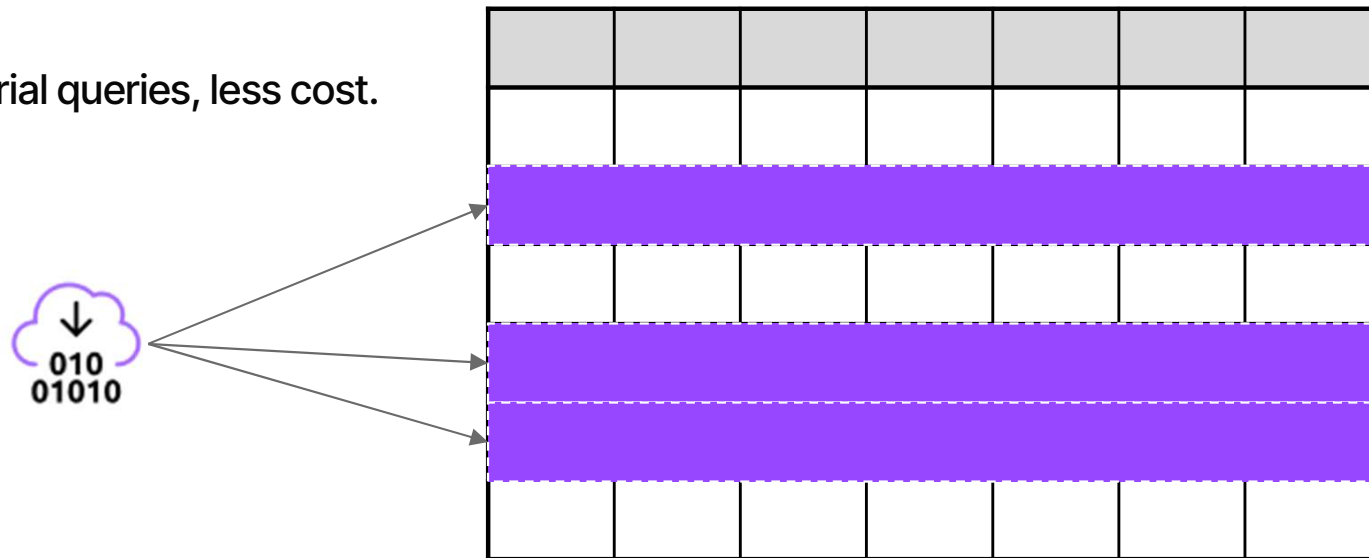
✗ `SELECT * FROM sales;`

✓ `SELECT customer_id, total FROM
sales WHERE region = 'EMEA';`



Tip #2: Use Limits and Sampling

- Use `LIMIT` when exploring data.
- Use sampling (`TABLESAMPLE` or random subsets) instead of querying billions of rows.
- **Benefits:** Faster trial queries, less cost.

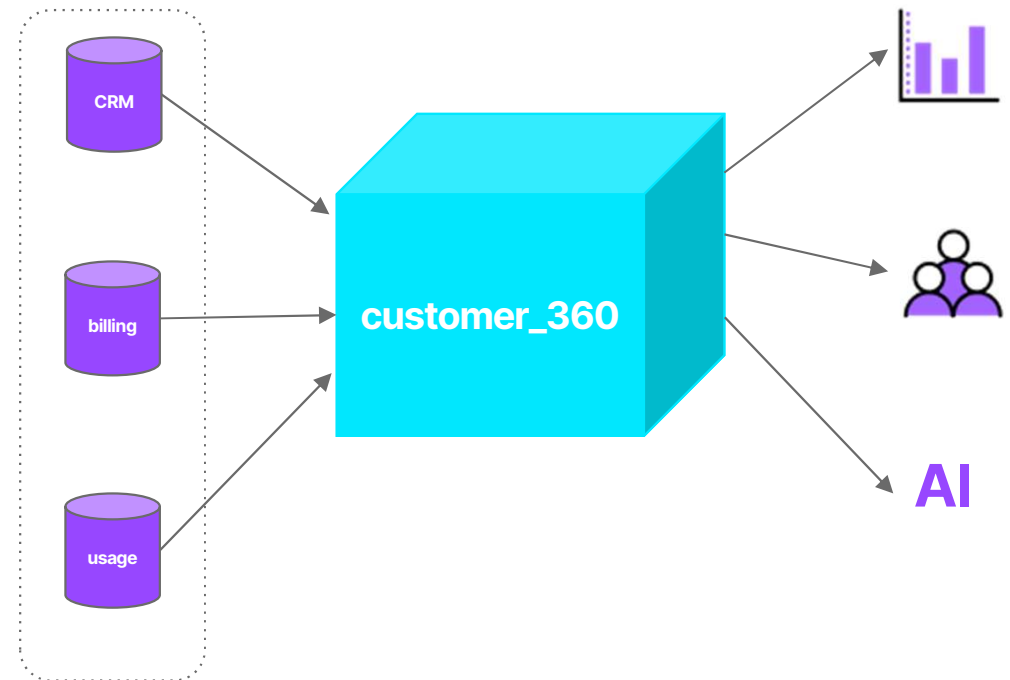


Tip #3: Leverage Curated Data Products

- Data products are pre-cleaned, business-ready datasets.
- Save time, improve trust, avoid duplication.

Example:

✅ Use `customer_360` instead of stitching together CRM + billing + usage manually



Common Pitfalls to Avoid

- ✗ Cartesian Joins: Forgetting join keys → exploding row counts.
- ✗ Unfiltered Queries: Querying entire tables unnecessarily.
- ✗ Too Many Columns: Selecting all fields instead of just what's needed.
- ✗ Ignoring Date/Partition Columns: Missing performance optimizations.



Group Activity

Review a slow query → Improve it together

Discuss how you would improve the following query:

```
SELECT * FROM sales JOIN  
customers;
```

Optimized query:

Instructor demonstration

Generating and Reviewing a
Query Plan (20 minutes)

Lesson summary

Performance & Best Practices

1. Why Performance Matters: Faster insights and lower costs are achieved with optimized queries, while slow queries waste time and money.
2. How Queries Execute: A Coordinator node plans queries and manages Worker nodes that execute tasks and process data. Starburst translates SQL into a query plan which shows data operations and potential bottlenecks.
3. Efficient Querying Tips:
 - a. Filter early: Select only necessary columns and use WHERE clauses.
 - b. Use limits and sampling: Employ LIMIT and TABLESAMPLE for faster, cheaper trial queries.
 - c. Leverage data products: Use pre-cleaned, business-ready datasets to save time and ensure trust.
4. Common Pitfalls to Avoid: Forgetting join keys (Cartesian Joins), unfiltered queries, selecting too many columns, and ignoring date/partition columns.